

Omkar Pathak

Staff Python Engineer · Technical Product Lead

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PROFESSIONAL SUMMARY

Staff-level Python engineer and cloud architect with 8 years of end-to-end product ownership – from ambiguous problem statement to production-grade systems. Deep expertise in GenAI, agentic workflows, ML, and cloud-native platforms. Obsessed with product thinking: every system I build starts with user empathy, clean UX, and measurable business impact. Proven track record of embedding AI into daily engineering workflows to dramatically accelerate delivery and quality. Solo-founded **TrackMyRupee**, a privacy-first financial product, owning the full stack from product design to live user feedback loops. Open-source author with **500k+ library downloads** globally.

CORE COMPETENCIES

Languages & Backend: Python (Advanced), SQL (PostgreSQL / MySQL / Snowflake), Bash, JavaScript; FastAPI, Django REST Framework, Flask, Spring Boot, NodeJS, Microservices, Serverless

AI / GenAI: LLMs (GPT-4, Llama), LangChain / LlamaIndex, RAG, Vector DBs (FAISS), Agentic Workflows, Prompt Engineering, Azure AI Foundry

Data & ML: Apache Airflow, Apache NiFi, Spark, Pandas, Scikit-learn, Informatica, Embulk, Google BigQuery

Frontend & Product: React, AngularJS, Bootstrap 5, Data Visualisation, UX-driven API design, end-to-end product ownership, solo founder experience

DevOps / Cloud / Security: Azure (Advanced), GCP, AWS, Docker / Kubernetes, CI/CD (Azure DevOps, GitHub Actions), OWASP, OAuth2 / JWT, SonarQube, PyTest

PROFESSIONAL EXPERIENCE

Senior Security Software Engineer

March 2026 – Present

ClearWater Security and Compliance

Pune, India

- Architected a Python-first security automation platform using serverless functions; designed automated incident response playbooks and ChatOps bots that substantially reduced mean time to respond (MTTR) – making security tooling intuitive and actionable for analyst teams.
- Led GenAI threat intelligence pipeline: built an end-to-end system that parses high-volume security alerts, extracts Indicators of Compromise (IoCs), and surfaces findings in a clean analyst-facing dashboard – reducing manual review workload significantly.
- Apply AI-assisted development workflows daily (LLM-assisted code review, automated test writing, alert triage) to compress engineering cycle time while maintaining OWASP and OAuth2 quality standards.
- Own all technical decisions across architecture, tooling selection, and security best practices for the engineering team.

Lead Consultant

May 2022 – March 2026

Emergys Solutions Pvt. Ltd.

Pune, India

Text-to-SQL AI Agent (Ellicium / Internal Product)

- Owned end-to-end: conceived the UX (natural language → data), architected a secure FastAPI + Serverless backend with JWT auth and rate limiting, and designed a multi-modal result experience (tables, summaries, charts) that let non-technical users self-serve analytics across all major RDBMS.
- RAG pipeline indexed schema metadata in FAISS; intent-driven table selection before SQL generation minimised hallucinations. Drove **INR 2.5 Cr+ (\$300K+)** in client revenue – **High Flyer Award Q4 2023 (top 1%)**.

Agentic Supplier Discovery (AlixPartners)

- Designed autonomous multi-agent pipeline on Azure AI Foundry (web scraping → entity extraction → qualification scoring → ranked UI) that cut a multi-day manual research task to minutes.

MMM, Supply Chain Risk & ECG Detection (AlixPartners / Ellicium)

- Built probabilistic MMM framework (Meta's Robyn) with interactive budget visualisations used by C-suite; led React + Python supply chain risk SaaS; shipped ECG anomaly detection algorithm via Flask API into a wearable device.
- Established Python coding standards, architectural review processes, and zero-downtime release protocols; used AI-assisted code review to enforce quality at scale.

Python ETL Developer

November 2020 – May 2022

Numerator

Pune, India

- Built Apache Airflow DAGs orchestrating ETL across PostgreSQL, Oracle, and Snowflake for an ad analytics product; authored ingestion pipelines from Google AdSense, Google Analytics, and Meta Ads Dashboard to surface unified spend-vs-performance views for marketing analysts.
- Built high-performance RESTful APIs and microservices using Django REST Framework for the Data Lake product suite; authored GitHub Actions + Docker CI/CD pipelines for automated deployments.

Data Engineer

July 2018 – November 2020

Schlumberger (SLB)

Pune, India

- Built containerised data ingestion framework using Apache NiFi and Google BigQuery – processed **1 TB of daily data** across **70+ global business units**; also solely developed **BigQuery DZ**, a cloud replacement for on-premise discovery-zone loading using Embulk with dynamic data-type mapping.
- Refactored Informatica migration pipelines with Azure DevOps CI/CD and custom Python validation – cut execution time from **9 hours to 1.5 hours** with near-zero manual errors.
- Built Spring Boot REST APIs powering an AngularJS ops dashboard for real-time ingestion job monitoring – designed for non-technical operations teams across all 70 business units.

OPEN SOURCE & PERSONAL PROJECTS

TrackMyRupee | *Python, Django ORM, Bootstrap 5, Docker, Sentry*

January 2026 – Present

- Solo product founder: designed UX, built full stack, shipped features, and iterated on real user feedback. Built **Smart Projections** (YTD savings → year-end forecast) with motivating UX and privacy-by-design architecture.

Pyresparser | *Python, NLP, SpaCy*

June 2018 – Present

- Resume parsing library using SpaCy NER; integrated into HR-tech products globally – **450k+ downloads** on PyPI/GitHub.

Pygorithm | *Python, Data Structures & Algorithms*

June 2016 – Present

- Educational algorithms library with **50k+ downloads**, featured as a top GitHub trending project.

EDUCATION

MMCOE, Savitribai Phule Pune University

June 2014 – May 2018

Bachelor of Engineering, Computer Science

Pune, India

HONOURS AND AWARDS

High Flyer Award | *Emergys Solutions*

Q4 2023

Awarded to top 1% of contributors across product and engineering teams for launching a GenAI SQL agent that drove **\$300K+** in client revenue and significantly enhanced internal analytics capabilities.

Quora Top Writer | *Quora*

2018

Recognised for technical thought leadership with **100K+ views** on Python, data engineering, and ML answers – widely referenced by students and junior developers globally.

Best Outgoing Student Award | *MMCOE*

2018

Selected from 180+ final-year engineering students for the highest combined performance in academics, innovation, and student leadership initiatives.